

```
241-0583.sql...d Raza (93) X
1  -- Q1
2  CREATE TRIGGER trg_after_insert_player
3  ON player
4  AFTER INSERT
5  AS
6  BEGIN
7      PRINT 'A new player has been added.';
8  END;
```

110 % 1 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:05:45.0310077-07:00

```
241-0583.sql...d Raza (93))
122 CREATE TRIGGER trg_audit_salary
123 ON player
124 AFTER UPDATE
125 AS
126 BEGIN
127     INSERT INTO audit_salary (pid, old_salary, new_salary)
128     SELECT i.pid, d.salary, i.salary
129     FROM inserted i
130     JOIN deleted d ON i.pid = d.pid
131     WHERE i.salary <> d.salary;
132 END;
133
134 -- Testing the triggers now
135 INSERT INTO player VALUES ('Test Player', 201, 22, 'Batsman', 50000, 1, NULL, NULL, NULL);
136
```

110 % 11 0 ↑ ↓

Messages

A new player has been added.

(1 row affected)

(1 row affected)

Completion time: 2026-04-03T10:49:14.5095840-07:00

Q2

```
24L-0583.sql...d Raza (93) X
1  -- Q1
2  CREATE TRIGGER trg_after_insert_player
3  ON player
4  AFTER INSERT
5  AS
6  BEGIN
7      PRINT 'A new player has been added.';
8  END;
9
10 -- Q2
11 CREATE TRIGGER trg_after_delete_team
12 ON team
13 AFTER DELETE
14 AS
15 BEGIN
16     DECLARE @name VARCHAR(25);
17     SELECT @name = tname FROM deleted;
18     PRINT 'Deleted team: ' + @name;
19 END;
```

110 % 2 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:07:32.1447686-07:00

```
24L-0583.sql...d Raza (93) X
122 CREATE TRIGGER trg_audit_salary
123 ON player
124 AFTER UPDATE
125 AS
126 BEGIN
127     INSERT INTO audit_salary (pid, old_salary, new_salary)
128     SELECT i.pid, d.salary, i.salary
129     FROM inserted i
130     JOIN deleted d ON i.pid = d.pid
131     WHERE i.salary <> d.salary;
132 END;
133
134 -- Testing the triggers now
135
136 -- Q1
137 INSERT INTO player VALUES ('Test Player', 201, 22, 'Batsman', 50000, 1, NULL, NULL, NULL);
138
139 -- Q2
140 INSERT INTO team VALUES ('Test Team', 99, 'Test Coach', 'Test City');
141 DELETE FROM team WHERE tid = 99;
```

110 % 11 0 ↑ ↓

Messages

(1 row affected)
Deleted team: Test Team

(0 rows affected)

(0 rows affected)

(1 row affected)

Completion time: 2026-04-03T10:49:55.0976932-07:00

Q3

```
24L-0583.sql...d Raza (93)
15 BEGIN
16     DECLARE @name VARCHAR(25);
17     SELECT @name = tname FROM deleted;
18     PRINT 'Deleted team: ' + @name;
19 END;
20
21 -- Q3
22 CREATE TRIGGER trg_after_update_salary
23 ON player
24 AFTER UPDATE
25 AS
26 BEGIN
27     DECLARE @old DECIMAL(10,2), @new DECIMAL(10,2);
28     SELECT @old = salary FROM deleted;
29     SELECT @new = salary FROM inserted;
30
31     IF ABS(@new - @old) > 5000
32         PRINT 'Salary changed by more than 5000.';
33 END;
```

110 % 2 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:08:36.6490080-07:00

```
24L-0583.sql...d Raza (93)
133
134 -- Testing the triggers now
135
136 -- Q1
137 INSERT INTO player VALUES ('Test Player', 201, 22, 'Batsman', 50000, 1, NULL, NULL, NULL);
138
139 -- Q2
140 INSERT INTO team VALUES ('Test Team', 99, 'Test Coach', 'Test City');
141 DELETE FROM team WHERE tid = 99;
142
143 -- Q3
144 -- Should trigger message (change is 10000)
145 UPDATE player SET salary = 100000 WHERE pid = 101;
146
147 -- Should NOT trigger message (change is only 1000)
148 UPDATE player SET salary = 101000 WHERE pid = 101;
```

110 % 11 0 ↑ ↓

Messages

Salary changed by more than 5000.

(1 row affected)

(1 row affected)

(1 row affected)

(1 row affected)

Completion time: 2026-04-03T10:50:21.8382530-07:00

Q4

```
24L-0583.sql...d Raza (93)
30
31 IF ABS(@new - @old) > 5000
32     PRINT 'Salary changed by more than 5000.';
33 END;
34
35 -- Q4
36 CREATE TRIGGER trg_insteadof_insert_player
37 ON player
38 INSTEAD OF INSERT
39 AS
40 BEGIN
41     IF EXISTS (SELECT 1 FROM inserted WHERE age < 18)
42         PRINT 'Insert blocked: player age is less than 18.';
43     ELSE
44         INSERT INTO player SELECT * FROM inserted;
45 END;
```

110 % 2 0

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:09:52.7996027-07:00

```
24L-0583.sql...d Raza (93)
140 INSERT INTO team VALUES ('Test Team', 99, 'Test Coach', 'Test City');
141 DELETE FROM team WHERE tid = 99;
142
143 -- Q3
144 -- Should trigger message (change is 10000)
145 UPDATE player SET salary = 100000 WHERE pid = 101;
146
147 -- Should NOT trigger message (change is only 1000)
148 UPDATE player SET salary = 101000 WHERE pid = 101;
149
150 -- Q4
151 -- Should be BLOCKED
152 INSERT INTO player VALUES ('Kid Player', 202, 15, 'Batsman', 30000, 1, NULL, NULL, NULL);
153
154 -- Should be ALLOWED
155 INSERT INTO player VALUES ('Adult Player', 203, 20, 'Batsman', 30000, 1, NULL, NULL, NULL);
```

110 % 11 0

Messages

Insert blocked: player age is less than 18.

(1 row affected)

A new player has been added.

(1 row affected)

(1 row affected)

Completion time: 2026-04-03T10:51:20.9789614-07:00

Q5

```

241-0583.sql...d Raza (93))
43     ELSE
44         INSERT INTO player SELECT * FROM inserted;
45     END;
46
47     -- Q5
48     ALTER TABLE player ADD total_runs INT DEFAULT 0;
49     ALTER TABLE player ADD total_wickets INT DEFAULT 0;
50
51     CREATE TRIGGER trg_after_insert_performance
52     ON performance
53     AFTER INSERT
54     AS
55     BEGIN
56         UPDATE player
57         SET total_runs = total_runs + i.runs,
58             total_wickets = total_wickets + i.wickets
59         FROM player p
60         JOIN inserted i ON p.pid = i.pid;
61     END;

```

110 % 5 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:19:20.6309225-07:00

```

241-0583.sql...d Raza (93))
153
154     -- Should be ALLOWED
155     INSERT INTO player VALUES ('Adult Player', 203, 20, 'Batsman', 30000, 1, NULL, NULL, NULL);
156
157     -- Q5
158     UPDATE player SET total_runs = 0, total_wickets = 0;
159     -- Check values before
160     SELECT pid, total_runs, total_wickets FROM player WHERE pid = 101;
161
162     -- Insert performance row
163     INSERT INTO performance VALUES (101, 203, 40, 1, 0);
164
165     -- Check values after - should increase by 40 runs, 1 wicket
166     SELECT pid, total_runs, total_wickets FROM player WHERE pid = 101;
167
168     -- Q6
169     -- add a temp team and player
170     INSERT INTO team VALUES ('Temp Team', 88, 'Temp Coach', 'Temp City');
171     INSERT INTO player VALUES ('Temp Player', 299, 25, 'Bowler', 40000, 88, NULL, NULL, NULL);
172
173     -- Verify player exists
174     SELECT * FROM player WHERE tid = 88;

```

110 % 11 0 ↑ ↓

Results Messages

	pid	total_runs	total_wickets
1	101	0	0

	pid	total_runs	total_wickets
1	101	40	1

Q6

```
24L-0583.sql...d Raza (93)
--
62
63 -- Q6
64 -- using INSTEAD OF because of the FK constraint
65 CREATE TRIGGER trg_insteadof_delete_team_players
66 ON team
67 INSTEAD OF DELETE
68 AS
69 BEGIN
70     -- First delete players of this team
71     DELETE FROM player
72     WHERE tid IN (SELECT tid FROM deleted);
73
74     -- Then delete the team itself
75     DELETE FROM team
76     WHERE tid IN (SELECT tid FROM deleted);
77 END;
78
79 -- Q7
80 CREATE TRIGGER trg_after_update_match_winner
81 ON match
82 AFTER UPDATE
83 AS
```

110 % 10 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T11:00:58.0073677-07:00

```
24L-0583.sql...d Raza (93)
--
170
171 -- Check values after - should increase by 40 runs, 1 wicket
172 SELECT pid, total_runs, total_wickets FROM player WHERE pid = 101;
173
174 -- Q6
175 INSERT INTO team VALUES ('Temp Team', 88, 'Temp Coach', 'Temp City');
176 INSERT INTO player VALUES ('Temp Player', 299, 25, 'Bowler', 40000, 88, NULL, NULL, NULL);
177
178 SELECT * FROM player WHERE tid = 88; -- should show Temp Player
179
180 DELETE FROM team WHERE tid = 88
181
182 SELECT * FROM player WHERE tid = 88; -- should return no rows
183
184 -- Q7
185 -- Should print message (as winner changes from 1 to 2)
186 UPDATE match SET winner_id = 2 WHERE mid = 201;
187
188 -- Should NOT print message (as winner stays the same)
189 UPDATE match SET venue = 'Multan' WHERE mid = 201;
190
191 -- Q8
```

110 % 10 0 ↑ ↓

Results Messages

	pname	pid	age	role	salary	tid	captain_id	total_runs	total_wickets
1	Temp Player	299	25	Bowler	40000.00	88	NULL	NULL	NULL

pname pid age role salary tid captain_id total_runs total_wickets

Q7

```
241-0583.sql...d Raza (93)
67 AS
68 BEGIN
69     DELETE FROM player
70     WHERE tid IN (SELECT tid FROM deleted);
71 END;
72
73 -- Q7
74 CREATE TRIGGER trg_after_update_match_winner
75 ON match
76 AFTER UPDATE
77 AS
78 BEGIN
79     IF EXISTS (
80         SELECT 1 FROM inserted i
81         JOIN deleted d ON i.mid = d.mid
82         WHERE i.winner_id <> d.winner_id
83     )
84         PRINT 'Match winner has been changed.';
85 END;
```

110 % 5 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:40:20.8702550-07:00

```
241-0583.sql...d Raza (93)
168 -- add a temp team and player
169 INSERT INTO team VALUES ('Temp Team', 88, 'Temp Coach', 'Temp City');
170 INSERT INTO player VALUES ('Temp Player', 299, 25, 'Bowler', 40000, 88, NULL, NULL, NULL);
171
172 -- Verify player exists
173 SELECT * FROM player WHERE tid = 88;
174
175 -- Delete team - should auto-delete the player too
176 DELETE FROM team WHERE tid = 88;
177
178 -- Should return no rows
179 SELECT * FROM player WHERE tid = 88;
180
181 -- Q7
182 -- Should print message (as winner changes from 1 to 2)
183 UPDATE match SET winner_id = 2 WHERE mid = 201;
184
185 -- Should NOT print message (as winner stays the same)
186 UPDATE match SET venue = 'Multan' WHERE mid = 201;
```

110 % 11 0 ↑ ↓

Messages

Match winner has been changed.

(1 row affected)

(1 row affected)

Completion time: 2026-04-03T10:54:32.7413962-07:00

Q8

```
241-0583.sql...d Raza (93) ✕
82         WHERE i.winner_id <> d.winner_id
83     )
84     PRINT 'Match winner has been changed.';
85 END;
86
87 -- Q8
88 CREATE TRIGGER trg_insteadof_delete_player
89 ON player
90 INSTEAD OF DELETE
91 AS
92 BEGIN
93     IF EXISTS (
94         SELECT 1 FROM performance
95         WHERE pid IN (SELECT pid FROM deleted)
96     )
97         PRINT 'Delete blocked: player has performance records.';
98     ELSE
99         DELETE FROM player WHERE pid IN (SELECT pid FROM deleted);
100 END;
```

110 % 5 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:43:48.4404434-07:00

```
241-0583.sql...d Raza (93) ✕
176 DELETE FROM team WHERE tid = 88;
177
178 -- Should return no rows
179 SELECT * FROM player WHERE tid = 88;
180
181 -- Q7
182 -- Should print message (as winner changes from 1 to 2)
183 UPDATE match SET winner_id = 2 WHERE mid = 201;
184
185 -- Should NOT print message (as winner stays the same)
186 UPDATE match SET venue = 'Multan' WHERE mid = 201;
187
188 -- Q8
189 -- Should be BLOCKED (pid 101 has performance records)
190 DELETE FROM player WHERE pid = 101;
191
192 -- Insert a player with no performance records, then delete - should be ALLOWED
193 INSERT INTO player VALUES ('No Stats Player', 301, 24, 'Bowler', 45000, 1, NULL, NULL, NULL);
194 DELETE FROM player WHERE pid = 301;
```

110 % 11 0 ↑ ↓

Messages

Delete blocked: player has performance records.

(1 row affected)

A new player has been added.

(1 row affected)

(1 row affected)

(1 row affected)

(1 row affected)

(1 row affected)

Completion time: 2026-04-03T10:55:40.2155575-07:00

Q9

```
24L-0583.sql...d Raza (93) X
--
94     SELECT 1 FROM performance
95     WHERE pid IN (SELECT pid FROM deleted)
96 )
97     PRINT 'Delete blocked: player has performance records.';
98 ELSE
99     DELETE FROM player WHERE pid IN (SELECT pid FROM deleted);
100 END;
101
102 -- Q9
103 CREATE TRIGGER trg_after_insert_perf_multirow
104 ON performance
105 AFTER INSERT
106 AS
107 BEGIN
108     DECLARE @total INT;
109     SELECT @total = SUM(runs) FROM inserted;
110     PRINT 'Total runs added in this operation: ' + CAST(@total AS VARCHAR);
111 END;
```

110 % 5 0 ↑ ↓

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:44:46.6449259-07:00

```
24L-0583.sql...d Raza (93) X
--
185 -- Should NOT print message (as winner stays the same)
186 UPDATE match SET venue = 'Multan' WHERE mid = 201;
187
188 -- Q8
189 -- Should be BLOCKED (pid 101 has performance records)
190 DELETE FROM player WHERE pid = 101;
191
192 -- Insert a player with no performance records, then delete - should be ALLOWED
193 INSERT INTO player VALUES ('No Stats Player', 301, 24, 'Bowler', 45000, 1, NULL, NULL, NULL);
194 DELETE FROM player WHERE pid = 301;
195
196 -- Q9
197 -- Single row insert - should print total runs = 30
198 INSERT INTO performance VALUES (106, 204, 30, 1, 0);
199
200 -- Multi-row insert - should print total runs = 90 (25+65)
201 INSERT INTO performance VALUES
202 (107, 205, 25, 0, 1),
203 (108, 202, 65, 0, 0);
```

110 % 11 0 ↑ ↓

Messages

(0 rows affected)

(1 row affected)
Total runs added in this operation: 30

(1 row affected)

(0 rows affected)

(2 rows affected)
Total runs added in this operation: 90

(2 rows affected)

Completion time: 2026-04-03T10:56:14.2848026-07:00

Q10

```

24L-0583.sql...d Raza (93)
112
113 -- Q10
114 CREATE TABLE audit_salary (
115     audit_id INT IDENTITY PRIMARY KEY,
116     pid INT,
117     old_salary DECIMAL(10,2),
118     new_salary DECIMAL(10,2),
119     changed_at DATETIME DEFAULT GETDATE()
120 );
121
122 CREATE TRIGGER trg_audit_salary
123 ON player
124 AFTER UPDATE
125 AS
126 BEGIN
127     INSERT INTO audit_salary (pid, old_salary, new_salary)
128     SELECT i.pid, d.salary, i.salary
129     FROM inserted i
130     JOIN deleted d ON i.pid = d.pid
131     WHERE i.salary <> d.salary;
132 END;

```

110 % 11 0

Messages

Commands completed successfully.

Completion time: 2026-04-03T10:46:54.5000007-07:00

```

24L-0583.sql...d Raza (93)
198 INSERT INTO performance VALUES (106, 204, 30, 1, 0);
199
200 -- Multi-row insert - should print total runs = 90 (25+65)
201 INSERT INTO performance VALUES
202 (107, 205, 25, 0, 1),
203 (108, 202, 65, 0, 0);
204
205 -- Q10
206 -- Update salary
207 UPDATE player SET salary = 95000 WHERE pid = 102;
208
209 -- Check audit table - should have 1 row with old/new salary and pid
210 SELECT * FROM audit_salary;
211
212 -- Update two players at once
213 UPDATE player SET salary = salary + 5000 WHERE tid = 1;
214
215 -- Audit table should now have more rows
216 SELECT * FROM audit_salary ORDER BY changed_at DESC;

```

110 % 11 0

Results Messages

	audit_id	pid	old_salary	new_salary	changed_at
1	1	101	90000.00	100000.00	2026-04-03 10:50:21.827
2	2	101	100000.00	101000.00	2026-04-03 10:50:21.827
3	3	102	85000.00	95000.00	2026-04-03 10:57:05.290

	audit_id	pid	old_salary	new_salary	changed_at
1	3	102	85000.00	95000.00	2026-04-03 10:57:05.290
2	4	203	30000.00	35000.00	2026-04-03 10:57:05.290
3	5	201	50000.00	55000.00	2026-04-03 10:57:05.290
4	6	102	95000.00	100000.00	2026-04-03 10:57:05.290
5	7	101	101000.00	106000.00	2026-04-03 10:57:05.290
6	1	101	90000.00	100000.00	2026-04-03 10:50:21.827
7	2	101	100000.00	101000.00	2026-04-03 10:50:21.827